Week #9: [Knee]

(Non-hyperlinked = Dr. Graham Briscoe Module)

Clinical Approach to the Knee: Part One

- Introduction to Module

Knee

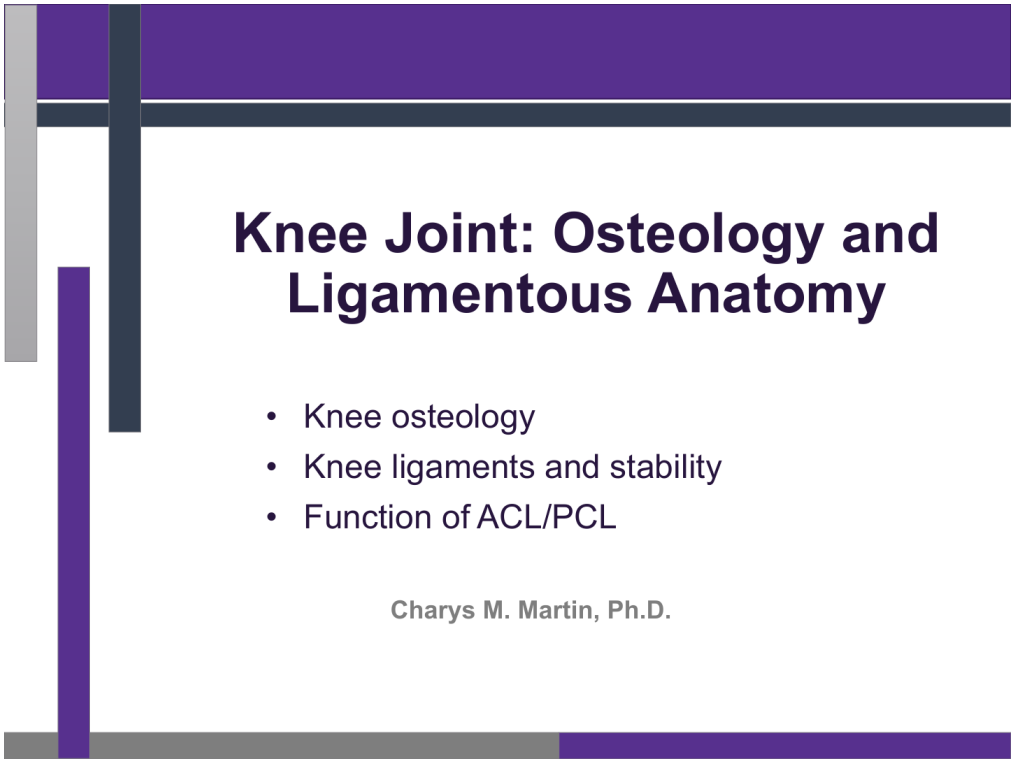
- Osteology and Ligamentous Anatomy (26 min)
- Ligamentous Injuries of the Knee (12 min)
- Meniscus Anatomy and Function (14 min)
- Meniscal Injuries (11 min)

Clinical Approach to the Knee: Part Two

- Introduction to Module

Knee

- Anterior Knee Pain Anatomy (5 min)
- Anterior Knee Pain Pathology (4 min)
- Miscellaneous Knee Pathology (8 min)
- Knee Injection Therapies (12 min)
- Clinical Approach to the Knee (9 min)



Knee Case #1 – Pivoting Injury

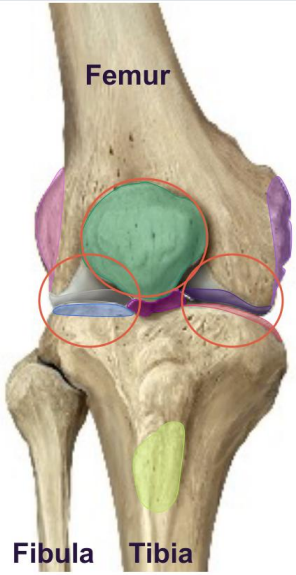
A 27 year old male football athlete pivots with his left knee as he changes direction to the right. He feels his knee "move" as it gives way. The athlete collapses to the ground and is unable to weight bear due to significant pain and swelling.



Osteology and Ligamentous Anatomy

- Knee osteology
- Knee ligaments and stability
- Function of ACL/PCL

Review Knee Joint: Osteology – Anterior View



Patella

Lateral Femoral Condyle

Medial Femoral Condyle

Medial Femoral Epicondyle

Lateral Femoral Epicondyle

Knee Joint

3 Articulations

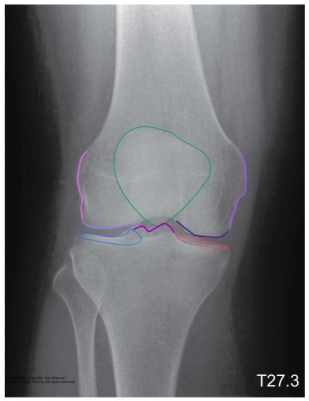
Tibial Tuberosity

Lateral Tibial Condyle

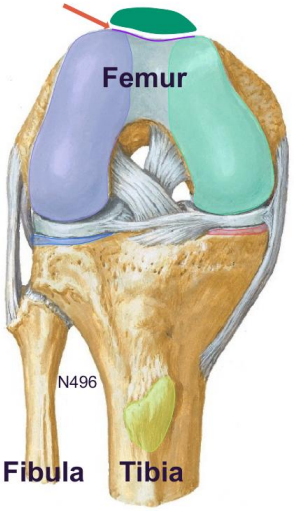
Medial Tibial Condyle

Intercondylar Eminence

AP View



Knee Joint: Osteology – Axial View



Patella

Lateral Femoral Condyle

Medial Femoral Condyle

Patellar Surface

Patellofemoral Joint

Tibial Tuberosity

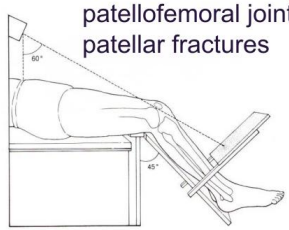
Lateral Tibial Condyle

Medial Tibial Condyle

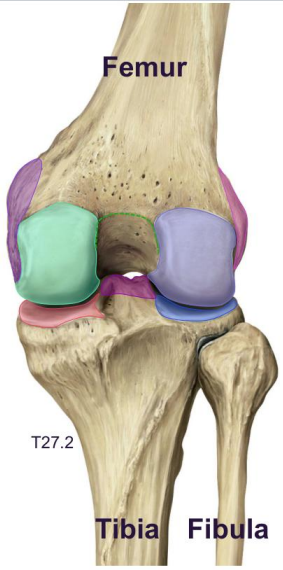


Skyline View

Good for assessing patellofemoral joint and patellar fractures



Knee Joint: Osteology – Posterior View



Patella

Lateral Femoral Condyle

Medial Femoral Condyle

Medial Femoral Epicondyle

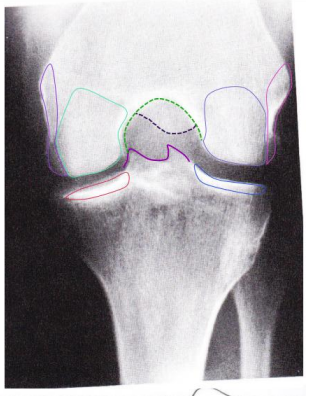
Lateral Femoral Epicondyle

Intercondylar Eminence

Lateral Tibial Condyle

Medial Tibial Condyle

Intercondylar Fossa (Notch)





Tunnel radiograph: good for seeing if any loose body or bone tissue in joint cavity

Knee Joint: Osteology – Lateral/Sagittal View



Patella

Lateral Femoral Condyle

Lateral Tibial Condyle

Tibial Tuberosity

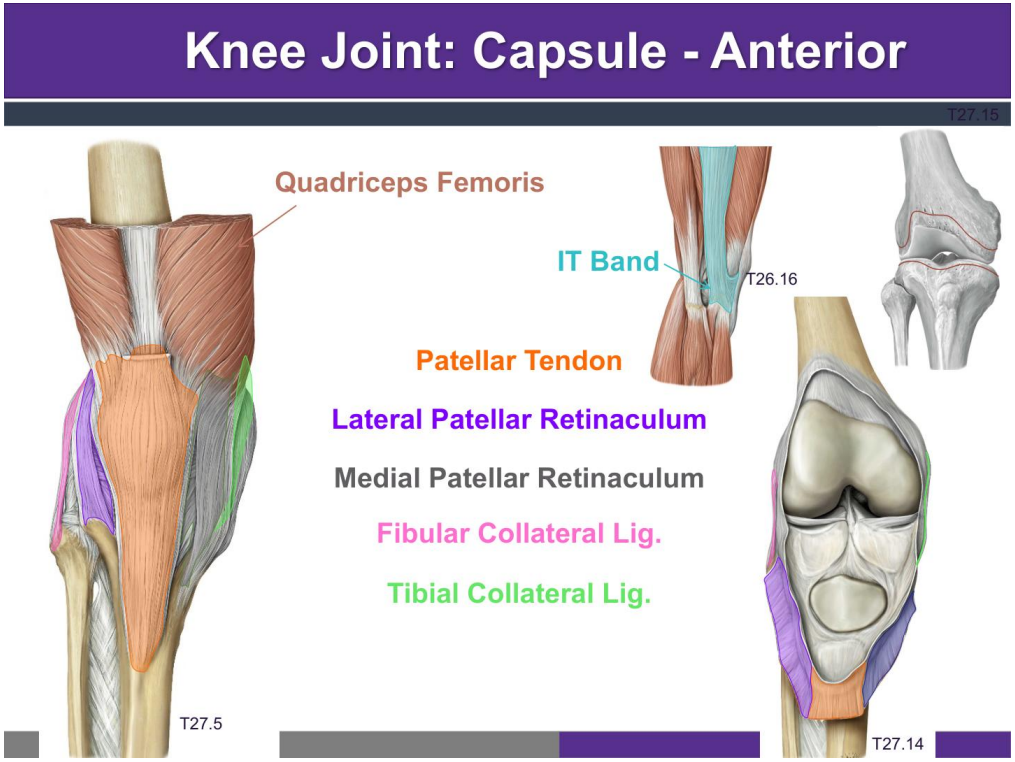


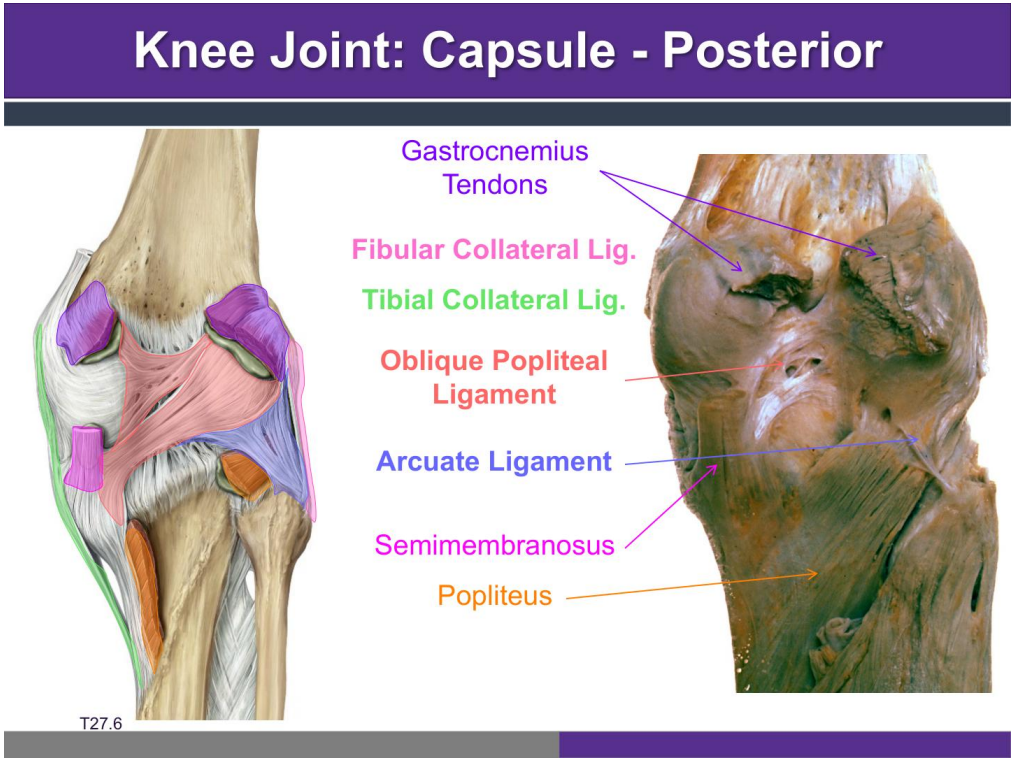
I. Knee Joint

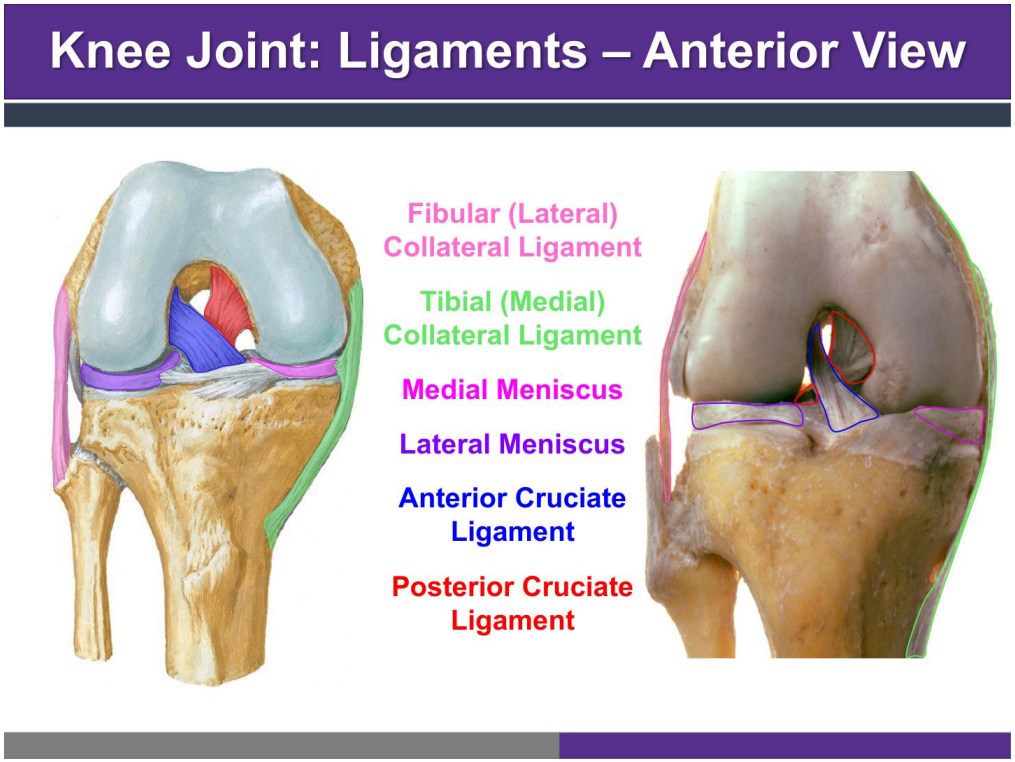
a. **Review:** osteology of the distal femur and proximal tibia
   i. ***Type:*** **Hinge** synovial joint; also allows for some **gliding/rotation** about the long axis
   ii. ***Articulation***: large **lateral and medial femoral condyles** articulate with **lateral and medial tibial condyles; femur** articulates with **patella**

Supporting Structures:

a. Stability is primarily facilitated by the ***surrounding musculature*** (especially the **quadriceps femoris**), with added stability conveyed by **ligamentous support**







Largest joint in the human body with the **largest joint capsule** and **extensive synovial membrane**

a. **Anteriorly:** the joint capsule is reinforced (and fibrous joint capsule is actually replaced) by the **tendon of the quadriceps femoris/patellar ligament** and by aponeurotic extensions of the vastus lateralis and the vastus medialis (medial and lateral patellar retinacula).
b. **Medially:** the joint capsule is reinforced by strong, long, and broad **tibial collateral ligament (TCL/MCL)** with muscular support from the **pes anserinus** (SGT), the **semimembranosus,** and the medial head of the **gastrocnemius**. Deep fibers of the TCL are firmly attached to the medial meniscus.

Clinical Note: TCL Injury: Frequency of TCL injuries are greater than fibular collateral ligament injuries. Common mechanisms include a lateral (valgus) force to an extended knee or excessive twisting/rotation of the flexed knee and because the medial meniscus is attached to the TCL, if the TCL is damaged the medial meniscus is also susceptible to damage.

- **TCL Assessment:** Testing of TCL integrity is performed by subjecting the flexed knee to a valgus stress (i.e. forced abduction of the leg at the femur), greater than normal abduction indicates TCL damage.

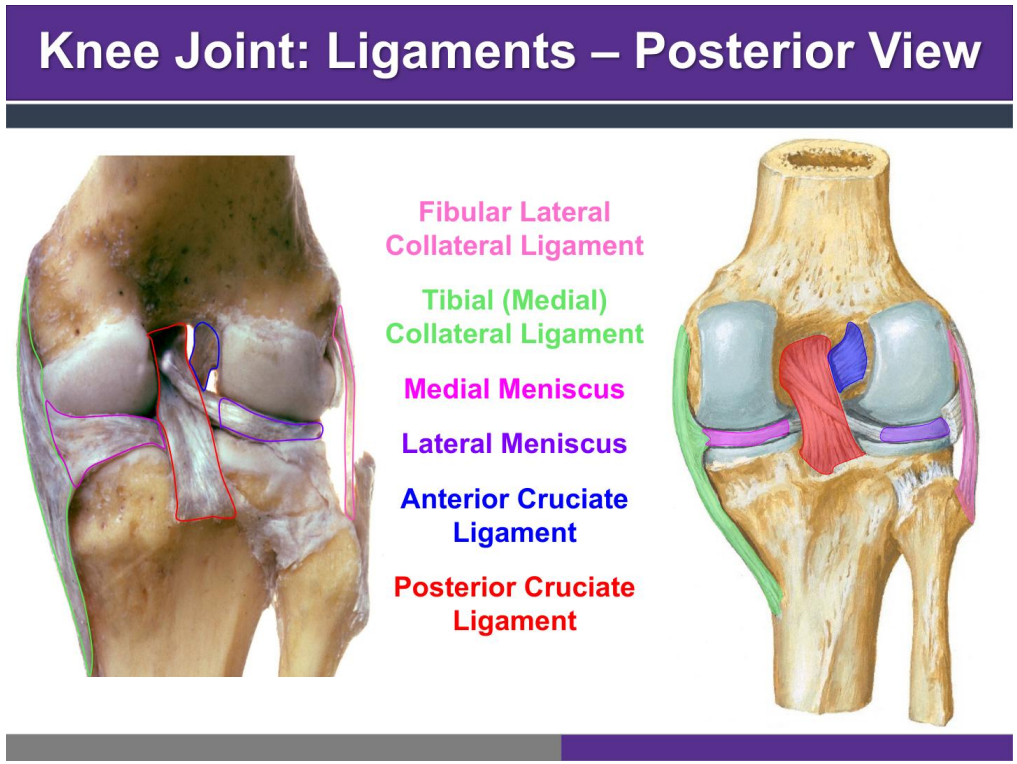
Laterally: joint capsule is reinforced by the strong, thin, and cord-like fibular collateral ligament (FCL/LCL) with musculotendinous support from the biceps femoris and the iliotibial band

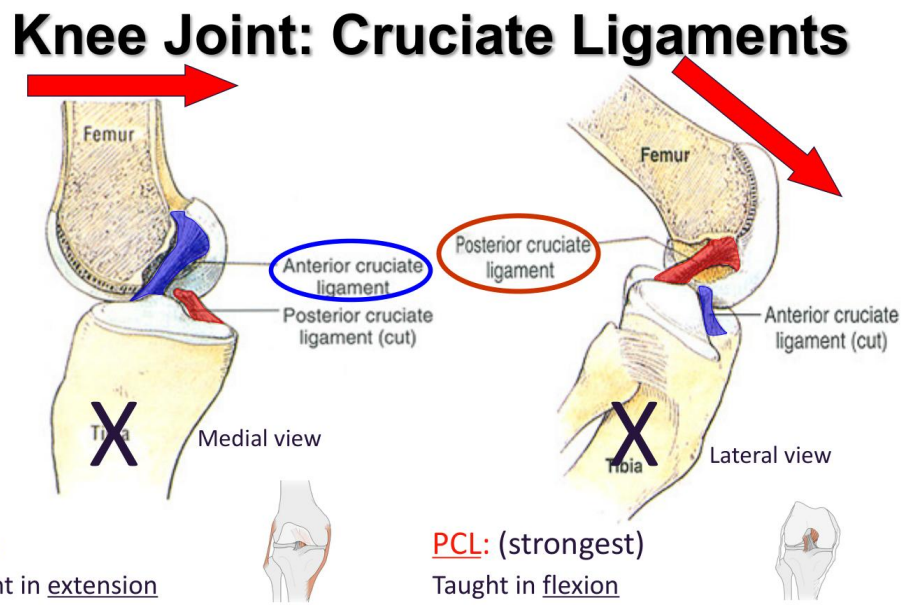
Clinical Note: FCL Injury: FCL can be damaged by a strong force applied on the medial aspect (varus) of the knee.

- **FCL Assessment:** Testing of the FCL is performed by subjecting the flexed knee to a varus stress (i.e. forced adduction of the leg at the femur), greater than normal adduction indicates FCL damage.

Posteriorly: the joint capsule covers the knee joint like a true fibrous capsule and is reinforce by the **oblique popliteal ligament**, which is a fibrous expansion from the **semimembranosus** tendon. The **arcuate popliteal ligament** arises from the posterior aspect of the fibular head, passes superomedially over the tendon of **popliteus,** and

spreads over the posterior surface of the knee.





ACL:

Taught in extension

Foot planted – prevents femur from moving post. on tibia

Foot free – prevents tibia from moving ant. under femur.

PCL: (strongest)

Taught in flexion

Foot planted – prevents femur from moving ant. on tibia

Foot free – prevents tibia moving post. under the femur.

e. Internally: the cruciate ligaments and the menisci are located within the joint capsule.

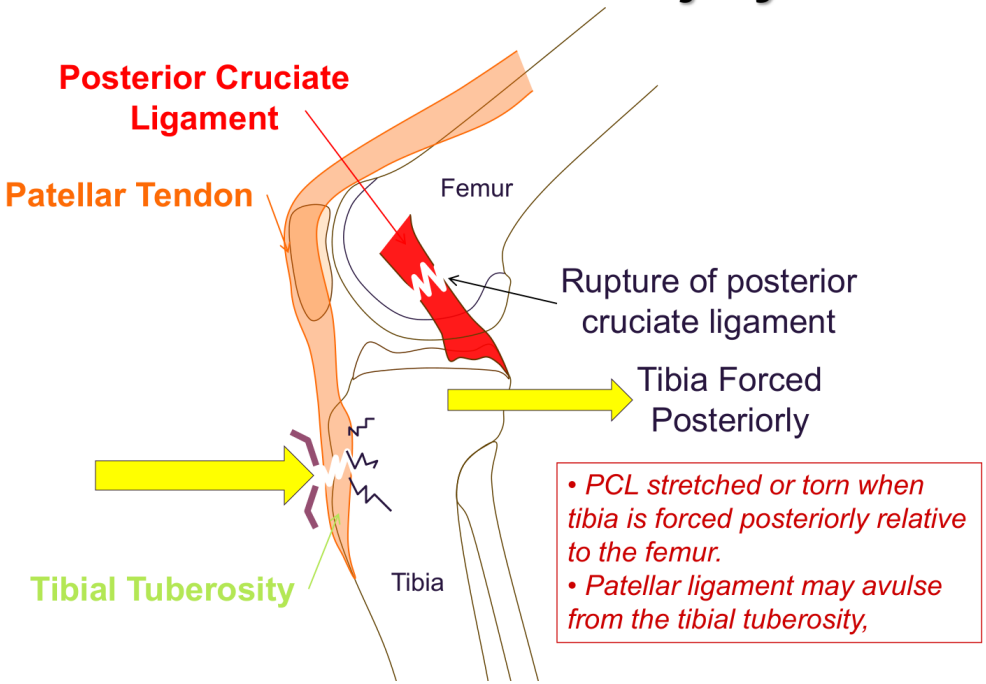
- **Note:** The cruciate ligaments are named for where they are attached to the tibial plateau.

Anterior cruciate ligament (ACL): thinner/weaker of the two cruciate ligaments; courses inferomedially from the medial aspect of the lateral femoral condyle to the intercondylar fossa of the tibia (just anterior to the intercondylar eminence); prevents excessive anterior translation of the tibia under the femur (prevents hyperextension of knee joint) and resists excessive medial rotation of tibia relative to femur

Clinical Note: ACL injury: common mechanism of injury involves receiving **forceful impact to thigh**, when the foot is fixed, forcing the tibia to translate anteriorly relative to femur. Can also be ruptured when a force applied to a flexed knee causes excessive rotation and medial (valgus) stress (causing the "**Unhappy triad**" of injuries, in which the ACL, TCL, and medial meniscus are all damaged)

- **ACL assessment:** Testing of the ACL involves the examiner assessing the extent/degree of anterior translation of the tibia under the femur. Greater than normal anterior translation of the tibia is indicative of an ACL injury.
  - **Lachman test:** supine patient flexes knee ~300 and examiner draws tibia forward; considered gold standard for ACL testing (ACL function is better isolated with this test)
  - **Anterior drawer test:** supine patient flexes hip 450 and knee 90 0; examiner palpates the hamstring tendons to ensure they are relaxed, before drawing tibia forward

PCL: Dashboard Injury



- *PCL stretched or torn when tibia is forced posteriorly relative to the femur.*
- *Patellar ligament may avulse from the tibial tuberosity,*

Posterior cruciate ligament (PCL): broader/stronger of the two cruciate ligaments; courses inferolaterally from the lateral side of the medial femoral condyle to the posterior aspect of the **intercondylar fossa** of the tibia; prevents **excessive posterior translation of the tibia on the femur**

Clinical Note: PCL injury: common mechanism of injury involves **'dashboard'** type event; an object strikes the anterior aspect of the tibia when the knee is in a flexed position, forcing the tibia to translate posteriorly relative to femur.

- **PCL assessment:** PCL integrity is tested via the ***posterior drawer test:*** similar to anterior drawer test, but examiner draws tibia posteriorly under the femur. Greater than normal posterior translation indicates PCL damage

Ligamentous Injuries of the Knee

- ACL and PCL injuries
- MCL and LCL injuries
- Evaluation, treatment, and return to play decisions

Jordan VanderEnde MD, FRCPC (PMR), CSCN EMG

- A 27 year old male football athlete pivots with his left knee as he changes direction to the right. He feels his knee "move" as it gives way. The athlete collapses to the ground and is unable to weight bear due to significant pain and swelling.



- Risk Factors
  - Female (4.5:1)
  - Pivoting sports
  - Anatomic and biomechanical factors
- Mechanism of Injury
  - Non-contact (60-80%) – cutting/pivoting during deceleration
  - Contact (30%) – direct blow causing hyperextension or valgus
- Presentation
  - "pop" from bone contusion
  - Immediate swelling / hemarthrosis
  - Sensation of "instability" or "giving out"
  - Pain lateral knee from bone contusion

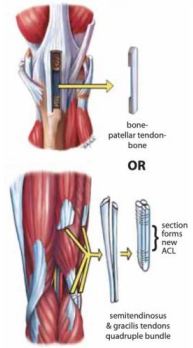






- Evaluation
  - History and physical examination
    - Lachman, Anterior Drawer, Pivot Shift
  - X-rays to rule out fractures, usually normal
  - MRI if unable to determine on exam
- Treatment
  - RICE, WBAT, crutches as needed
  - Physiotherapy
  - Ortho referral for reconstruction
- Return to Play
  - 9-12 months after surgery
    - 3-6% reinjury rate

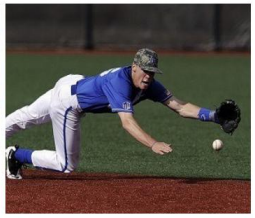




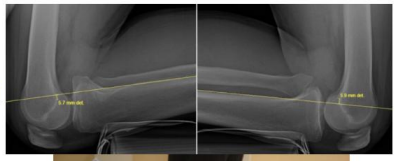
Anterior Cruciate Ligament Tear. Chapter 27 – Acute Knee Injuries. Brukner and Khan, Clinical Sports Medicine 3E, McGraw-Hill Professional.

- Mechanism of injury
  - Direct blow to anterior tibia with knee flexed
  - Hyperextension
- Presentation
  - Poorly defined pain, sometimes posteriorly
  - Minimal swelling
  - Less common than ACL tears
  - Also associated with meniscal and chondral injuries



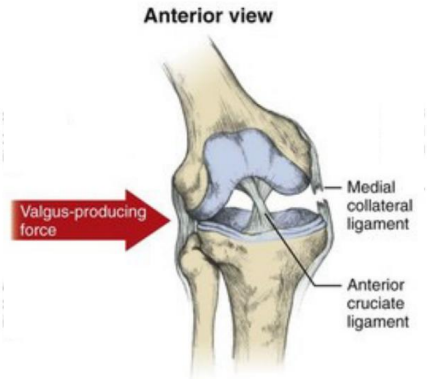


- Evaluation
  - History and physical examination
    - Posterior drawer test, sag sign, reverse Lachman
  - X-rays and stress x-rays (grading)
  - MRI if suspected PLC injury
- Treatment
  - Physiotherapy generally sufficient
  - PCL bracing
  - Surgical reconstruction if rotatory instability or with posterolateral corner injuries
- Return to play
  - 50% return to same or higher level of activity





- Very common
- Almost always caused by collisions
- Valgus force
- Swelling (minimal to none)
- Sometimes associated with medial meniscal tears
- Injury can range from sprain to complete tear
- Complete tear is often associated with ACL tear
- **Treatment**: early range of motion, bracing



- Uncommon in isolation
- Almost always caused by collisions
- Varus force (high energy)
- No swelling in isolation
- Associated with PCL injuries
- Injury can range from sprain to complete tear
- **Treatment**: bracing, surgical repair if other injuries and/or instability



- **Medial Collateral Ligament (MCL)**





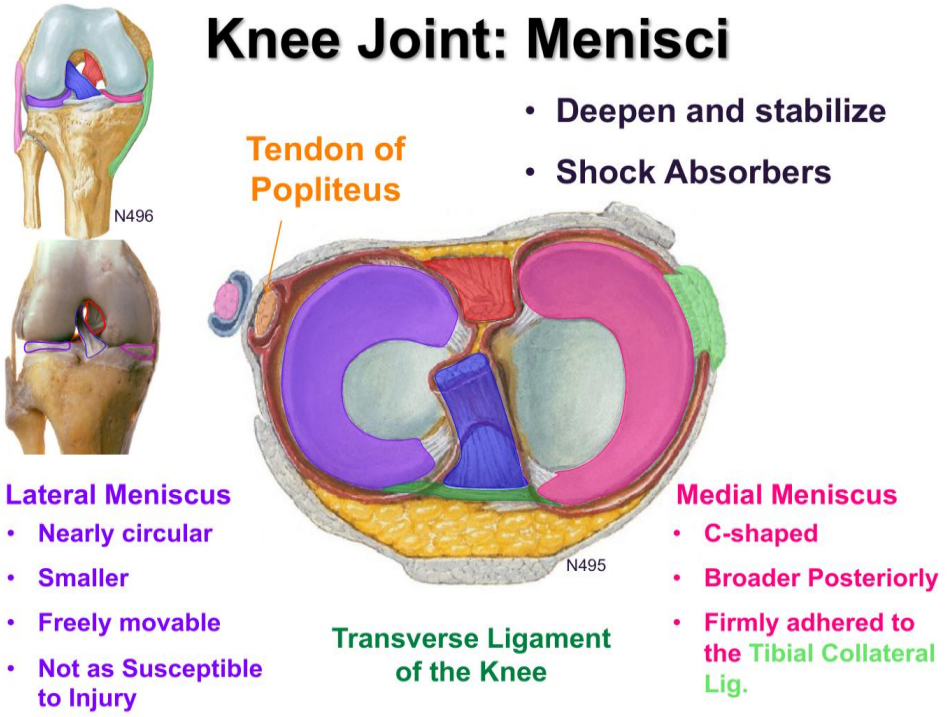
- **Lateral Collateral Ligament (LCL)**



Ligamentous Injuries of the Knee (ACL and PCL injuries, MCL and LCL injuries, Evaluation, treatment, and return to play decisions) ***No text notes from Dr. Briscoe**

Meniscus Anatomy and Function

- Anatomy and function of meniscus
- Bucket-handle tears
- Locking mechanism of the meniscus



iii. Medial meniscus: C-shaped, broader posteriorly than anteriorly, limited mobility due to attachment to deep fibers of the TCL; increases the congruency for the articulating surface of the femoral condyles and plays a role in shock absorption

Clinical Note: the medial meniscus is the more likely to be torn/damaged than the lateral meniscus due to its **lack of mobility** and attachment to the **TCL; usually damaged** during stresses to the knee that also ***affect the TCL***, including the twisting/valgus mechanism that results in the **'Unhappy triad'** set of injuries

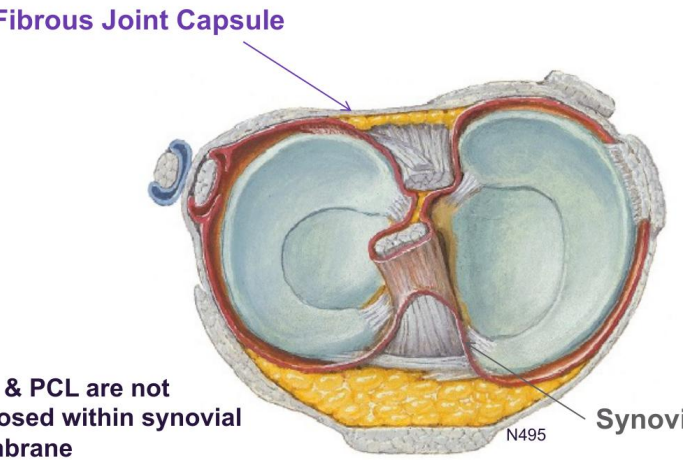
iv. Lateral meniscus: More **circular, smaller,** and **more mobile** of the menisci; increases the congruency for the articulating surface of the femoral condyles and plays a role in shock absorption

Meniscus Anatomy and Function

- Anatomy and function of meniscus
- Bucket-handle tears
- Locking mechanism of the meniscus

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Knee Joint: Capsule & Synovium



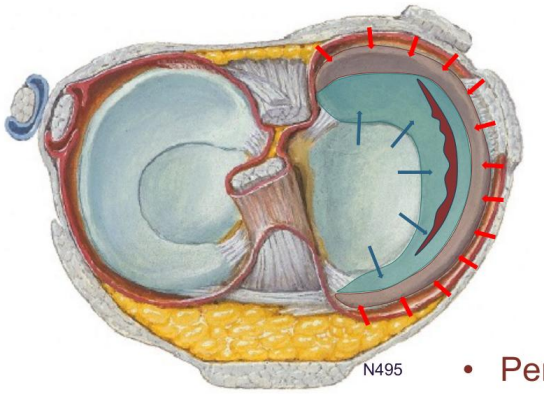
- **ACL & PCL are not enclosed within synovial membrane**

Knee Case #2 – Twisting Injury

A 23 year old male ice hockey athlete falls awkwardly to the ice while his skate is caught up with another player. His knee rotates and he suffers immediate medial pain. There is no obvious initial swelling but develops a moderate amount of swelling later on that night.



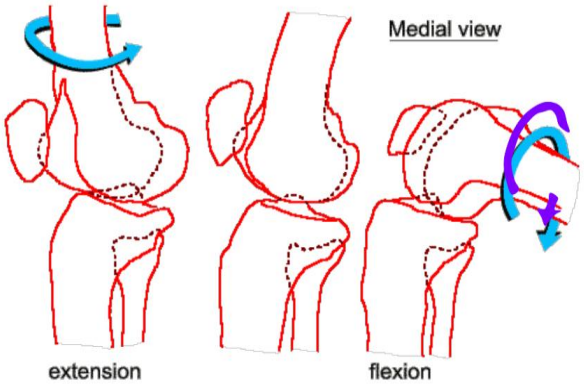
Medial Meniscus: Arterial Supply



- Peripheral meniscus richly vascularized
- Central meniscus is avascular; nourished by synovial fluid

Knee Joint: Locking Mechanism

Femur rotates medially on tibia during the final 15 to 20° of extension



***Screw Home Mechanism**: Medial rotation maximally tightens the ligaments of the knee*



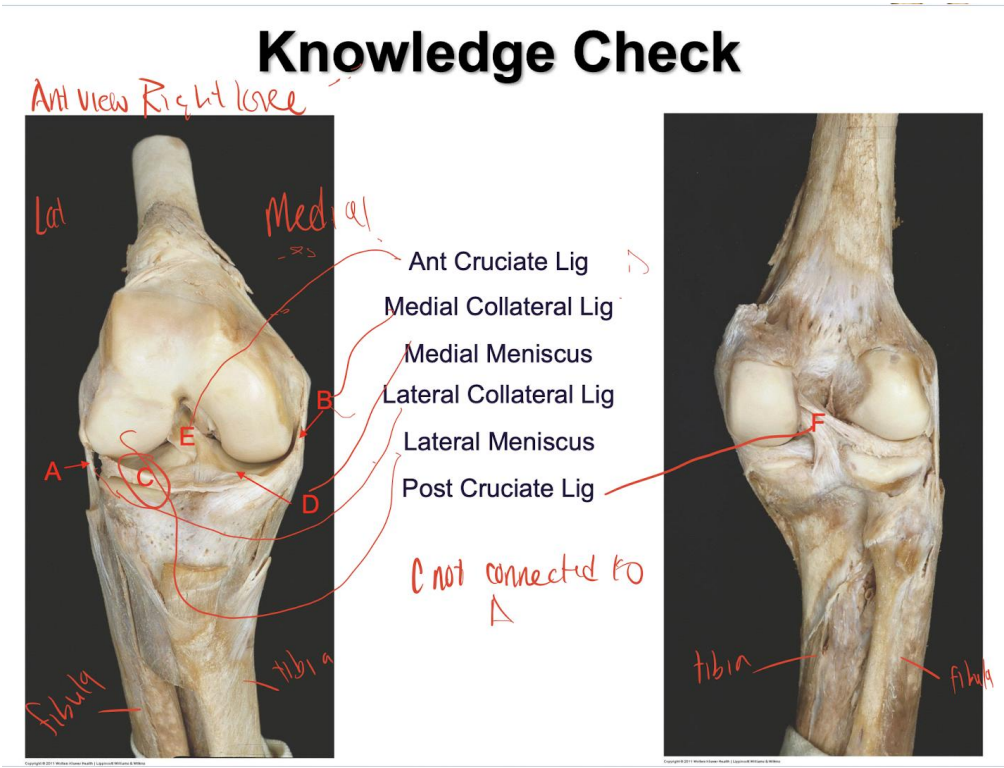
Popliteus contracts to rotate the femur 5° "unlocking the knee" for flexion to occur



Knee Extension: the shapes of condyles and menisci cause medial rotation of the femur on the tibia

Knee Flexion: Popliteus "unlocks" the knee joint by laterally rotating the femur

Knowledge Check



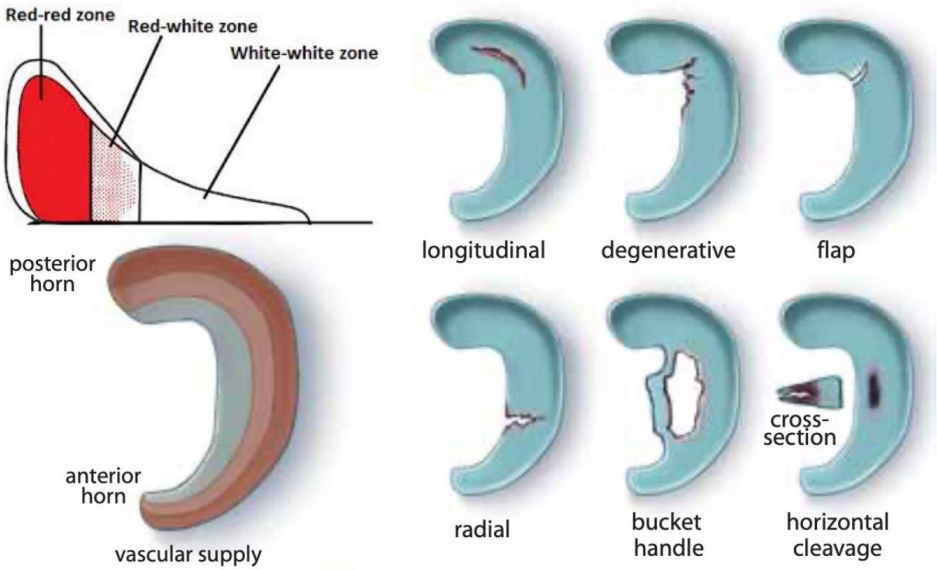
Meniscal Injuries

- Pathology, mechanism of injury, symptoms
- Treatment of meniscal injuries
- Unhappy triad

Jordan VanderEnde MD, FRCPC (PMR), CSCN EMG

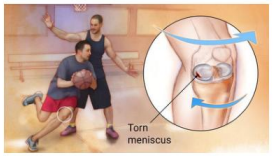
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Meniscal Injuries. Chapter 27 – Acute Knee Injuries. Brukner and Khan, Clinical Sports Medicine 3E, McGraw-Hill Professional.

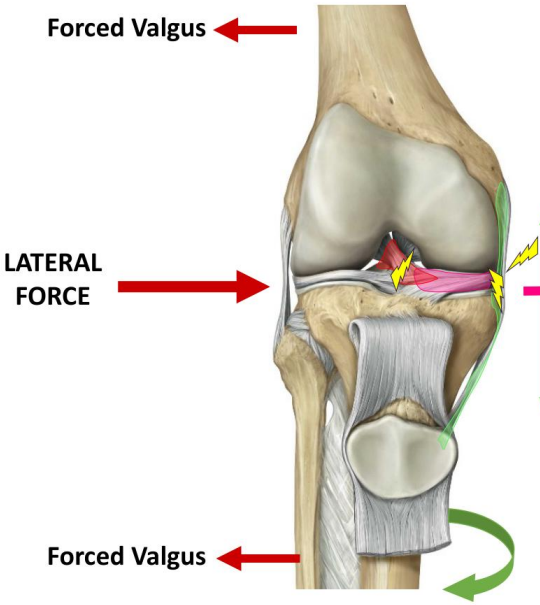
- Mechanism of Injury
  - Variable based on age/degenerative changes
  - Rotation or twisting with foot planted
- Presentation
  - Medial or lateral joint line pain following injury (or delayed)
  - Delayed or intermittent swelling/effusions
  - +/- mechanical symptoms (locking and catching)
- Evaluation
  - History and physical examination
    - Joint effusion & joint line tenderness, McMurray, Thessaly
  - X-rays (usually normal, but can assess for degree of OA)
  - MRI if failing conservative treatment or suspected large tear



- Treatment
  - Physiotherapy: maintain/improve ROM, strengthen quads/hamstrings to improve stability
    - Active vs passive physio
  - Medications: oral OTC, corticosteroid intra-articular injection
  - Bracing/activity modifications: accommodative
  - Surgery: depends on presence and severity of OA
    - Arthroscopy: meniscal repair vs debridement
    - Arthroplasty: last resort

- Treatment
  - Physiotherapy: maintain/improve ROM, strengthen quads/hamstrings to improve stability
    - Active vs passive physio
  - Medications: oral OTC, corticosteroid intra-articular injection??
  - Bracing/activity modifications: accommodative??
  - Operative repairs
    - Meniscal repair: red or red/white, larger tears, bucket handle, meniscal root tears, younger patients
    - Meniscal resection: white, chronic degenerative
    - Partial meniscectomy, meniscal repair, meniscal transplantation
- Return to play
  - Variable depending on meniscectomy versus repair
  - 6 weeks to 3 months

Unhappy Triad



1. *forced valgus stretches MCL – ligament tears from joint*
2. *bucket-handle tear of the medial meniscus since the ligament is attached to MCL*

3. forced abduction, external rotation and slight flexion causes a tear in the ACL

Not mentioned in text notes

Meniscus Injuries (Pathology, mechanism of injury, symptoms, Treatment of meniscal injuries, Unhappy triad) ***No text notes from Dr. Briscoe**

Anterior Knee Pain Anatomy

- Anatomy of patellofemoral joint
- Stability forces of patellofemoral joint

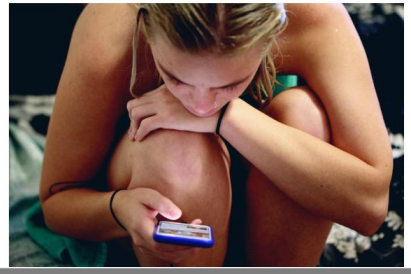
Anterior Knee Pain Anatomy

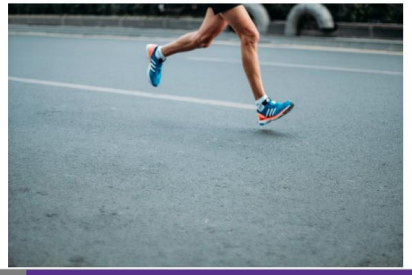
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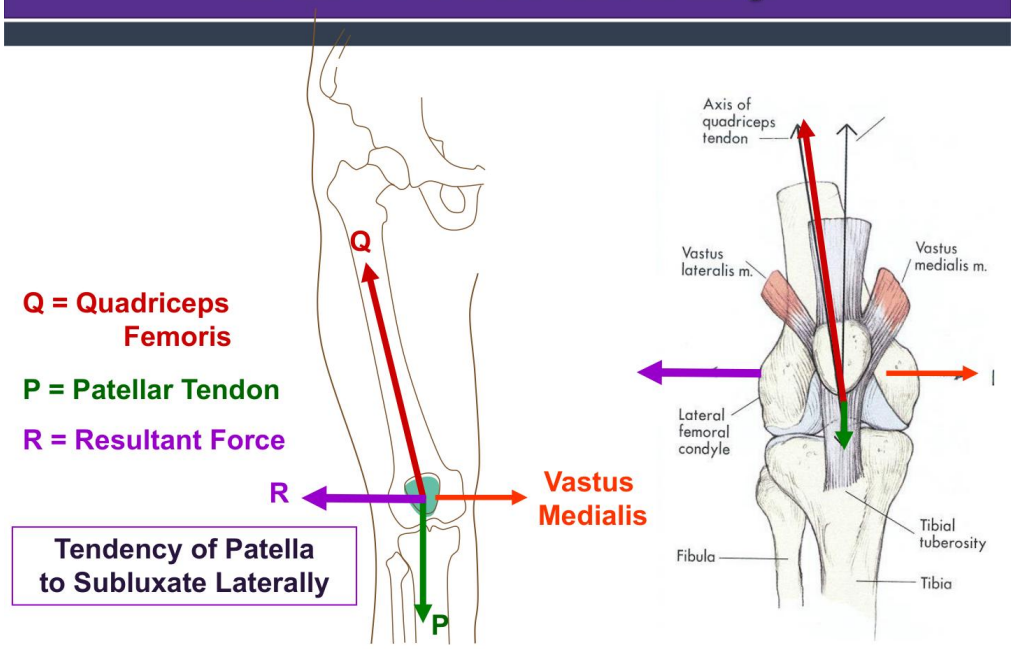
Knee Case #3 – Anterior Knee Pain

A 18 year old female presents with a three month history of worsening bilateral anterior knee pain. There was no injury but she recently started running. Her pain was initially only with exercise but now it is painful with prolonged sitting and stairs. There is no swelling and she describes the pain as a deep anterior aching pain.





Knee Joint: Stability



Q = Quadriceps Femoris

P = Patellar Tendon

R = Resultant Force

Tendency of Patella to Subluxate Laterally

I. **Knee Joint**

Review: osteology of the distal femur and proximal tibia

- *Type:* **Hinge** synovial joint; also allows for some **gliding/rotation** about the long axis
- *Articulation:* large **lateral and medial femoral condyles** articulate with **lateral and medial tibial condyles; femur** articulates with **patella**
- *Supporting Structures:*

1. Stability is primarily facilitated by the **surrounding musculature** (especially the **quadriceps femoris**), with added stability conveyed by **ligamentous support**
2. **Largest** joint in the human body with the **largest joint capsule** and **extensive synovial membrane**
   a. Anteriorly: the joint capsule is reinforced (and fibrous joint capsule is actually replaced) by the **tendon of the quadriceps femoris/patellar ligament** and by aponeurotic extensions of the vastus lateralis and the vastus medialis (**medial** and **lateral patellar retinacula**).
   b. Medially: the joint capsule is reinforced by strong, long, and broad **tibial collateral**

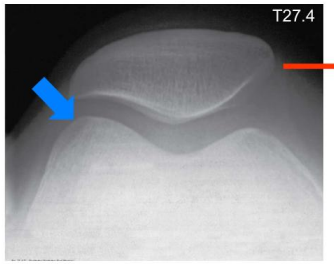
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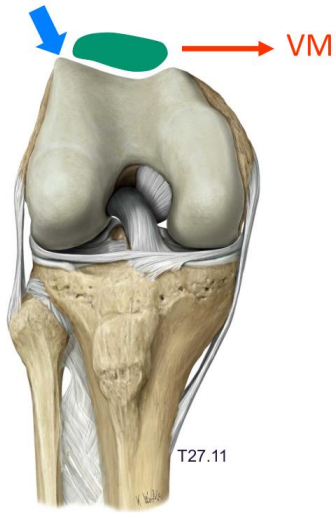
Knee Joint: Stability

Two mechanisms counteracting the tendency for the patella to subluxate laterally during knee extension:

1. Pull by Vastus Medialis

1. Extra buttress of bone on lateral patellar surface





Clinical Note: Knee Deformities: In normal anatomical position, the femur is placed diagonally within the thigh. This creates a Q-angle (the angle between a line drawn from ASIS to patella and another line drawn through the patella and tibial tuberosity). Normal Q-angles result in normal distribution of weight between the medial and lateral femoral/tibial condyles.

- **Genu Varum (Bowlegged):** The femur is angled more vertically; thus, decreasing the Q-angle
  - Increases the weight-bearing load on the medial femoral/tibial condyles
- **Genu Valgus (Knock-kneed):** The femur is at a dramatic angle; thus, increasing the Q-angle
  - Increases the weight-bearing load on the lateral femoral/tibial condyles
- As a result of the Q-angle the quadriceps pull the patella Superolaterally, which has a tendency to subluxate the patella laterally. There are two mechanisms to resist this lateral pull on the patella: the **vastus medialis** and an extra buttress of bone on the lateral patellar surface of the femur.

Anterior Knee Pain Pathology (Patellofemoral Pain Syndrome (PFPS) - Symptoms, Presentation, Treatment) ***No text notes from Dr. Briscoe**

MISC Knee Structures

- Anatomy of Knee Bursae

laterally. There are two mechanisms to resist this lateral pull on the patella: the **vastus medialis** and an extra buttress of bone on the lateral patellar surface of the femur.

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- Increases the weight-bearing load on the medial femoral/tibial condyles

Genu Valgus (Knock-kneed): The femur is at a dramatic angle; thus, increasing the Q-angle

- Increases the weight-bearing load on the lateral femoral/tibial

- *Movements:* Flexion/extension; some minor degree of rotation when knee is flexed
- *Associated Structures:* There are many **bursae** located around the knee that function to protect all of the tendons that are crossing the knee joint from rubbing against the bone
  a. **Prepatellar bursa:** Located superficial to the patella allows the skin to move freely during knee movement
  b. **Infrapatellar bursa:** Superficial and deep bursa that lie superficial and deep to the patellar ligament. Allow the skin and patellar ligament to move freely during knee movements.
  c. **Suprapatellar Bursa:** Lies deep to the quadriceps femoris tendon and attaches to a small muscle, articularis genu, and the synovial membrane. When the quadriceps extend the knee articularis genu will pull the bursa tight to allow the quadriceps femoris tendon and patella to move freely over the femur.
  ❖ ***Clinical Note:*** Irritation or swelling of a bursa due to chronic friction is known as **bursitis**. **Housemaid's knee** is usually caused by friction between the skin and the patella, affecting the **prepatellar bursa**. Commonly seen in individuals who move around and work on their knees often. **Clergyman's knee** is usually caused by excessive friction between the skin and the tibial tuberosity, affecting the **superficial infrapatellar bursa**. Commonly seen in individuals who sit on their knees on hard surfaces for extended periods of time.

b. *Innervation:* Hilton's Law: sensory innervation to a joint is the same as the nerve supply to muscles that act upon the joint. Knee: Femoral nerve and sciatic nerve; also articular branches from obturator nerve and saphenous nerve (due to proximity)
c. *Blood supply:* **genicular arterial anastomoses** formed by four branches of the **popliteal artery**: **superior and inferior medial genicular arteries and superior and inferior lateral genicular arteries.** The descending branch of the lateral circumflex femoral artery provides collateral circulation to the knee.

Note: While the **popliteus** does not add much posterior stability to the joint, it is important for:
a. unlocking the knee from terminal extension (initiates knee flexion); and
b. pulling the lateral meniscus posteriorly during knee flexion

- During natural extension the shapes of the femoral/tibial condyles and the menisci cause the femur to rotate medially on the tibia
- This rotation allows the joint to be more stable, enabling 'quiet standing' (limited muscular contraction to maintain stance).
- The popliteus muscle contracts to laterally rotate the femur, to 'unlock the knee', to allow flexion of the knee.

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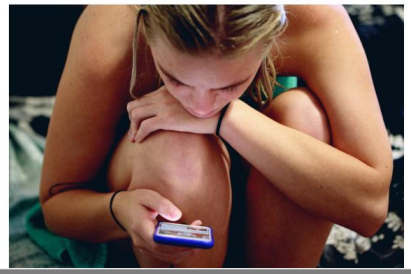
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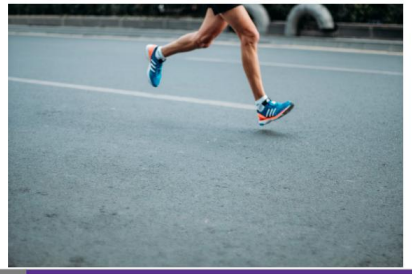
MISC Knee Pathology (IT band friction syndrome, Popliteal (Baker's) cyst) ***No text notes from Dr. Briscoe**
Knee Injection Therapies (Corticosteroids, Hyaluronic acid, Platelet Rich Plasma (PRP), Mesenchymal stem cells) ***No text notes from Dr. Briscoe**
Clinical Approach to the Knee (History, Approach to traumatic and atraumatic pain, Imaging tests) ***No text notes from Dr. Briscoe**



Knee Case #3 – Anterior Knee Pain

- A 18 year old female presents with a three month history of worsening bilateral anterior knee pain. There was no injury but she recently started running. Her pain was initially only with exercise but now it is painful with prolonged sitting and stairs. There is no swelling and she describes the pain as a deep anterior aching pain.

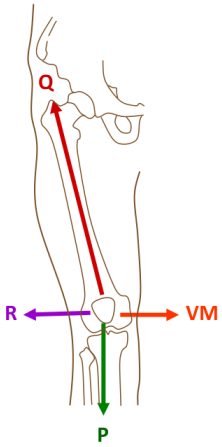




Patellofemoral Pain Syndrome (PFPS)

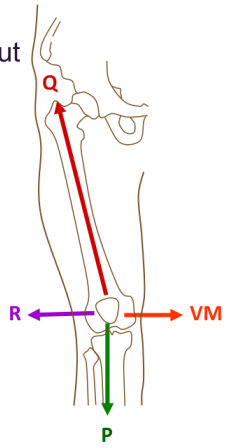
- Clinical syndrome of anterior knee pain during or after repetitive loading of extensor mechanism

- Risk factors
  - Females, adolescents
  - Increased tibio-femoral angle and Q-angle
- Presentation
  - Very common (30% lifetime prevalence)
  - Gradual onset (biomechanical overload)
  - Traumatic (patellar contusion)
  - Anterior knee pain, worse after activity and prolonged sitting, aching pain
  - No joint swelling*



Patellofemoral Pain Syndrome (PFPS)

- Evaluation
  - Clinical diagnosis: history and physical examination
  - No imaging tests needed
    - Sometimes x-rays and MRIs are ordered to rule out other pathologies if failing to respond to treatment
- Treatment
  - Activity modifications
  - Physiotherapy
    - Dynamic misalignment
    - Balance of vastus lateralis and vastus medialis
  - Bracing or taping
- Return to play

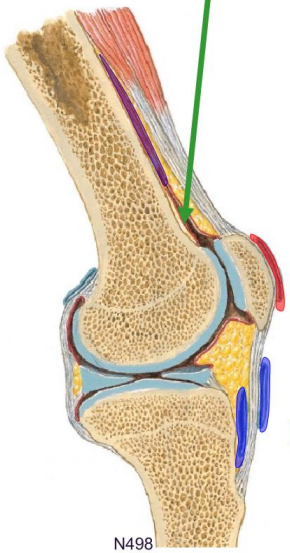


MISC Knee Pathology

- Patellar bursitis
- IT band friction syndrome
- Popliteal (Baker's) cyst

Jordan VanderEnde MD, FRCPC (PMR), CSCN EMG

Knee Bursitis



- Common problem in the knee
- Often caused by chronic irritation from friction
- Clinical diagnosis: mild swelling and tenderness over bursa
- Treatment: PRICE, NSAIDs
- Don't confuse with: septic bursitis, trauma/associated injuries

N498

LO #3

Popliteal (Baker's) Cyst





- A cystic extension of the synovial cavity/bursae (gastrocnemius/semimembranosus) into the popliteal fossa through defects in the posterior fibrous joint capsule
- Adults:
  - Uncomfortable swelling in the popliteal fossa
  - May limit flexion
  - Can be large and extend distal to mid-calf
- Investigations:
  - US (confirm cyst), X-ray (look for orthopaedic injuries), MRI?
- Differential:
  - Popliteal artery aneurysm, soft tissue tumour, DVT, meniscal cyst
- Treatment: cons. for 6 weeks* unless vascular/neural compromise
  - PRICE, NSAIDs, ?IA corticosteroid injection
  - If not responding -> MRI, surgical referral (not just a cyst removal)

IT Band Friction Syndrome

- Friction between ITB and lateral epicondyle during stance phase (friction occurs below 30° of knee flexion)

- Mechanism of injury
  - Running (especially downhill)
  - Friction less likely with sprinting
- Presentation
  - Aching lateral knee during running
- Treatment
  - PRICE, stretch ITB



IT Band Friction Syndrome. Chapter 29 – Lateral, Medial and Posterior Knee Pain. Brukner and Khan, Clinical Sports Medicine 3E, McGraw-Hill Professional.

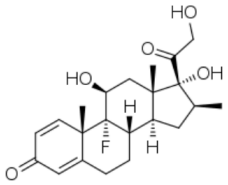
Knee Injection Therapies

- Corticosteriods
- Hyaluronic acid
- Platelet Rich Plasma (PRP)
- Mesenchymal stem cells

Jordan VanderEnde MD, FRCPC (PMR), CSCN EMG

Corticosteroids

- Corticosteroid (aka “steroid” or “cortisone”) injections
  - Local delivery of a potent anti-inflammatory
  - Used for treatment of joint, bursa and tendon disorders
- Benefits
  - Rapid onset
  - Inexpensive (<$20)
- Limitations
  - Short duration of action (<4-12 weeks)
  - evidence of chondrotoxicity and progression of OA if injected intra-articularly *(weak evidence)**





*Effect of Intra-articular Triamcinolone vs Saline on Knee Cartilage Volume and Pain in Patients With Knee Osteoarthritis: A Randomized Clinical Trial. McAlindon TE, LaValley MP, Harvey WF, Price LL, Driban JB, Zhang M, Ward RJ. JAMA. 2017 May 16;317(19):1967-1975. doi: 10.1001/jama.2017.5283. PMID: 28510679

Hyaluronic Acid (HA)

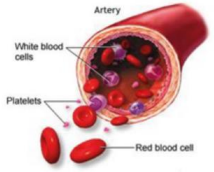
- Hyaluronic Acid (HA) Injections
  - FDA approval in 2001 for intra-articular use
  - HA is found intrinsically in synovial fluid
    - Replacing HA theoretically increases viscoelasticity, increasing shock absorption and lubrication of joint
- Benefits
  - >26 weeks of relief
  - Good safety profile
- Limitations
  - Expensive: $300-$500
  - Heterogeneous studies



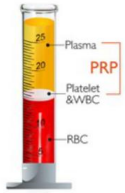
Sports Med. 2017 May;47(5):807-828. doi: 10.1007/s40279-016-0620-z. Biological Therapies in Regenerative Sports Medicine. Andia I[1], Maffulli N

Platelet-Rich Plasma (PRP)

- Platelet-rich plasma (PRP) injections
  - Autologous preparation of patient's whole blood (20-60 mL)
    - Centrifuged or filtered (3-10 mL) and injected into area of interest
    - Supraphysiologic concentration of platelets, growth factors
  - Used for treatment of muscle, joint, tendon and ligament injuries
- Potential benefits
  - Improve the repair and/or regeneration of tissues
- Limitations
  - Generally considered safe
  - Expensive: $500-$1,500
  - Studies with high risk of bias
  - Lack of standardization of PRP



[Blood components]



[After centrifugation]

Sports Med. 2017 May;47(5):807-828. doi: 10.1007/s40279-016-0620-z. Biological Therapies in Regenerative Sports Medicine. Andia I[1], Maffulli N

PRP – Evidence



- Cochrane Database (2014)[1]
  - 19 RCTs (1,088 patients)
  - PRP vs. placebo for:
    - Soft tissue injuries: lateral elbow tendinopathy, patellar tendinopathy and Achilles tendinopathy
    - Adjunct to surgery: ACL reconstruction, rotator cuff repair, subacromial decompression and Achilles tendon rupture repair
  - Results
    - High risk of bias, lack of standardization of PRP preparation
    - Overall: **no clinically significant improvement in pain and function**
- IOC Consensus on PRP (2010)[2]
  - "Firm recommendations on the effectiveness of PRP in the clinical setting to support the healing processes of muscle, tendon, ligament and cartilage injuries **cannot be given**"

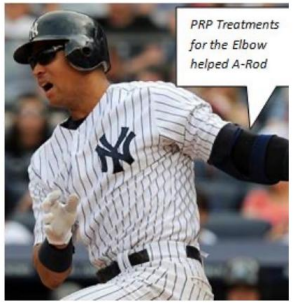
1.) Moraes VY, Lenza M, Tamaoki MJ, Faloppa F, Belloti JC. Platelet-rich therapies for musculoskeletal soft tissue injuries. Cochrane Database Syst Rev 2014;4:CD010071.24782334
2.) Br J Sports Med. 2010 Dec;44(15):1072-81. doi: 10.1136/bjsm.2010.079822. IOC consensus paper on the use of platelet-rich plasma in sports medicine.

But PRP Helped these Athletes….

- Tiger Woods
  - PRP for Achilles tendon injury, ACL repair



- Alex Rodriguez
  - 5 PRP injections post hip surgery (2009), and lateral elbow tendinopathy

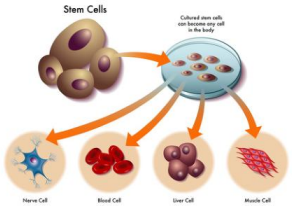


- Kobe Bryant
  - PRP for knee OA



Mesenchymal stem cells (MSCs)

- Mesenchymal stem cells (MSCs)
  - Potential for articular cartilage regeneration? Not shown
- Benefits
  - Not shown to be better than other inj.
- Limitations
  - Very limited human clinical studies
  - ++expensive ($3,000-$10,000)







Mesenchymal stem cells (MSCs)

- As per May 15th, 2019
  - Health Canada announced that it considers stem cell therapies to be a form of drug treatment, which means that it must be formally approved after undergoing rigorous review

Ottawa

Crackdown on unproven stem cell therapies long overdue, Ottawa scientist says

Dr. Michael Rudnicki says provinces need to follow Health Canada's lead

CBC News - Posted: Jul 10, 2019 4:00 AM ET | Last Updated: July 10, 2019

Health · **Second Opinion**

'Stem cell' therapies offered at private clinics need to be approved as drugs, Health Canada says

Health · **Second Opinion**

A stem cell 'cautionary tale' as Health Canada cracks down on private clinics

- **Confirm that you have ceased the sale and advertisement of ADSC until such time as:**
  - **You receive a No Objection Letter (NOL) pursuant to a Clinical Trial Application (CTA) for investigational use of the product, or**
  - **You receive Notice of Compliance (NOC) and Drug Identification Number (DIN) for the product in question as evidence of market authorization, your company obtains an Establishment Licence (EL), and your company meets all other applicable requirements related to the importation and sale of drug products.**

Your formal response to Health Canada related to this matter is requested by Friday, May 24, 2019.

Page 1 of 4

Canada

This an excerpt from one of the letters Health Canada sent to 36 private clinics across the country ordering them to stop advertising and selling cellular therapies. (CBC News)

Clinical Approach to the Knee

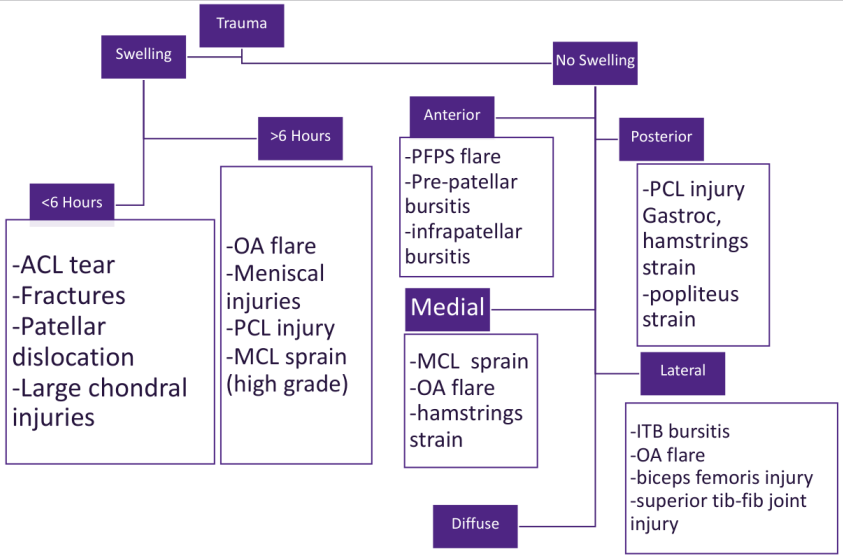
- History
- Approach to traumatic and atraumatic pain
- Imaging tests

Jordan VanderEnde MD, FRCPC (PMR), CSCN EMG

Clinical Approach – History

- Patient Symptoms
  - Pain (aching, burning, throbbing, stabbing)
  - Inflammatory (swelling, stiffness, redness, warmth)
  - Neurologic (tingling, shooting, numbness, heaviness)
- Patient Function
  - Structural (instability, catching, giving way, **locking**)
  - Pain/neurologic (weakness, limping, trouble walking)
- Provoking Factors
  - Weight bearing (walking, standing, running)
  - Stairs
  - Sitting
  - Inactivity

Clinical Approach – Traumatic Knee Pain



Non-Traumatic Knee Pain

Anterior Pain	Posterior Pain	Medial Pain	Lateral Pain
PFPS	PCL injury	Medial meniscal injury	IT band friction syndrome
Patellar tendinopathy	Popliteus tendinopathy	Medial knee OA	Biceps femoris tendinopathy
Quadriceps tendinopathy	Gascrocnemius tendinopathy	Pes anserine bursitis	Lateral meniscal injury
Infrapatellar bursitis	Baker's cyst	MCL injuries	Peroneal nerve injury
Pre-patellar bursitis	Biceps femoris tendinopathy	Referred pain from hip	Lateral knee OA
Apophysitis	DVT	Radiculopathy	Referred pain from hip
Stress fracture of patella	Neoplasm		Radiculopathy
Referred pain from Hip	Referred pain from hip		Neoplasm
Neoplasm	Radiculopathy		Superior tib-fib joint pathology

References

- Chapter 29 – Lateral, Medial and Posterior Knee Pain. Brukner and Khan, Clinical Sports Medicine 3E, McGraw-Hill Professional.
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